



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Total Boundness and Compactness

ϵ -net and Total Boundness

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CONDUCTED BY

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ϵ -net and Total Boundness

ϵ -net

Let (X, d) be a metric space and $\epsilon > 0$. A subset $N \subseteq X$ is called an ϵ -net for X if for every point $x \in X$, there exists a point $n \in N$ such that $d(x, n) < \epsilon$. In other words, the collection of open balls $\{B(n, \epsilon) : n \in N\}$ covers the entire space X .

② Example of ϵ -net

Show that the set $Z = \{-3, -2, -1, 0, 1, 2, 3\}$ is a 1-net for $\mathbb{R}$.



② Example of ϵ -net

The set $B = \{B(m, n) : m, n \in \mathbb{Z}\}$ is a $\sqrt{2}$ -net for $\mathbb{R}^2$.



Total Boundness

A metric space (X, d) is said to be *totally bounded* if for every $\epsilon > 0$, there exists a finite ϵ -net for X . In other words, for any given $\epsilon > 0$, we can find a finite set of points in X such that the open balls centered at those points with radius ϵ cover the entire space X .

Example of Total Boundness

The interval $[0, 1]$ is totally bounded.



② Example of Total Boundedness

The set $\mathbb{R}$ is not totally bounded.



② Example of Total Boundedness

The infinite discrete space $\mathbb{N}$ with the discrete metric is not totally bounded.



② Example of Total Boundedness

The infinite-dimensional space ℓ^2 is not totally bounded.



Separability and Compactness

💡 **Totally Boundedness and Separability**

Every totally bounded metric space is separable.



💡 **Compactness and Separability**

Every compact metric space is separable.



Total Boundness and Compactness

Total Boundness and Cauchy Subsequences

A metric space (X, d) is totally bounded if and only if every sequence in X contains a Cauchy subsequence. 

Sequential Compactness and Total Boundness

A metric space (X, d) is sequentially compact if and only if it is complete and totally bounded. 

Lebesgue number and Lebesgue Covering Lemma

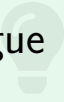
Lebesgue Number

Let $\{G_\alpha : \alpha \in \Lambda\}$ be an open cover of a metric space (X, d) . A real number $\lambda > 0$ is said to be a *Lebesgue number* for the open cover $\{G_\alpha\}$ if for each subset A of X with $d(A) < \lambda$, there is at least one set G_α which contains A .

In other words, any subset of X that is "small enough" (in terms of diameter) can be entirely contained within one of the sets in the cover $\mathcal{U}$.

Lebesgue Covering Lemma

Every open cover of a sequentially compact metric space has a Lebesgue number.



Sequential Compactness and Compactness

💡 In MS, Sequential Compactness $\iff$ Compactness

A metric space (X, d) is sequentially compact if and only if it is compact. 💡

Equivalent Characterizations of Compactness

💡 Equivalent Characterizations of Compactness

For a metric space (X, d) , the following statements are equivalent: 

1. X is compact.
2. X is sequentially compact.
3. X is complete and totally bounded.
4. Bolzano-Weierstrass property

Thank You!

We'd love your questions and feedback.

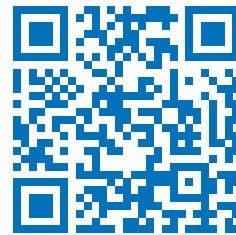
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(Lectures, walkthroughs, and course updates)



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References

- [1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.
- [2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.